

JINALKUMARI VANIYA

New Hyde Park, NY, 11040 | 516 424 4018 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Hofstra University

Master of Science in Data Science (GPA: 3.9)

Jan 2025 – May 2027

Vishwakarma Government Engineering College

Bachelor of Engineering (B.E.) in Information Technology

July 2015 – June 2019

Technical skills

Programming Languages: Python, SQL, R

Machine Learning & AI: Scikit-learn, PyTorch, Deep Learning, NLP, LLMs, LangChain, RAG, OpenWebUI

Data & Visualization: Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn

Tools & Platforms: PostgreSQL, MongoDB, ChromaDB, FAISS, Snowflake, Azure, AWS, Docker, Streamlit, Git

Work experience

AI Analyst | Hotwire Communications Ltd. | New York, USA

April 2026 – Present

- Identified and resolved inconsistencies between AI-driven WiFi automation rules and compliance requirements
- Analyzed automation rule behavior to diagnose failures and improve system reliability
- Maintaining and enhancing an internal HR RAG chatbot (Ask Toby) using OpenWebUI and ChromaDB to deliver accurate, context-aware responses

Research Assistant | Hofstra University | New York, USA

Oct 2025 – March 2026

- Developed LangChain pipelines to generate and store 100+ structured outputs for analysis
- Analyzed 900+ responses using NLP techniques (NLTK, POS tagging) to identify linguistic and demographic biases
- Evaluated model behavior across 25+ categories using N-shot prompting and reproducible evaluation workflows

Projects

Clinical RAG Chatbot | Python, LangChain, FAISS, LLMs, Streamlit | AI/ML Project ([GitHub](#))

- Built a RAG-based clinical chatbot for context-aware, evidence-grounded medical responses.
- Developed a vector search pipeline for semantic retrieval across medical documents.
- Integrated LLM generation with source citations for transparent, grounded responses.

Knee Osteoarthritis (OA) Severity Grading | Python, PyTorch, Swin Transformer, ResNet, Streamlit | Capstone Project ([GitHub](#))

- Developed a multimodal deep learning system combining knee X-rays and clinical metadata for osteoarthritis severity classification.
- Evaluated 9 deep learning architectures and applied ordinal classification and class-imbalance techniques, improving macro F1-score by ~20% over the strongest image-only baseline.
- Built an interactive Streamlit application for real-time severity prediction, interpretation, and clinical assessment.

READMIN – ICU Discharge Risk & Intervention Bot | Python, Streamlit, Scikit-learn, SHAP ([GitHub](#))

- Built a machine learning system for ICU discharge risk prediction, incorporating SHAP-based explainability for patient-level risk analysis.
- Developed clinician- and patient-facing Streamlit interfaces for risk assessment, workflow monitoring, and intervention tracking.
- Designed structured JSON workflows and logging pipelines to support system integration, traceability, and audit monitoring

Data Warehouse Project: Advanced Portfolio Analysis | Python, SQL, Scikit-learn, NLP, Trello, Llama ([GitHub](#))

- Built a portfolio analysis system using SQL and MongoDB pipelines to process and analyze structured financial data.
- Applied transformer-based embeddings and NLP to identify relationships between financial assets and generate portfolio insights.
- Developed scalable analytical workflows for trend analysis, semantic retrieval, and portfolio performance evaluation.

Hospital Emergency Room Dashboard | Excel, Power BI, SQL ([GitHub](#))

- Built an interactive Dashboard using Power Query and DAX to clean data and track key hospital metrics.
- Visualized patient visits, wait times, admissions, referrals, and satisfaction scores to help identify trends and support decision-making.

Certifications

- Microsoft Azure AI Essentials Professional Certificate** — Microsoft & LinkedIn
- Microsoft Azure Essentials Professional Certificate** — Microsoft & LinkedIn
- Generative AI Professional Certificate** — Snowflake
- Docker Foundations Professional Certificate** — Docker